

Time dependent muscle energetics: PCr and recovery heat

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1 Introduction

This document details a mathematical model for the time-dependent response of the energetic recovery to skeletal muscle contraction. The modelling within the document contains an ODE describing the time-dependent response of the muscle fibre to ATP consumption. First the model is described and parameters are determined. Second, the model is applied to an experimental condition from [BAG95], where muscle was repeatedly contracted and the initial and recovery heats were recorded.

2 Methods

2.1 ATP Regeneration

Given the complexities of ATP regeneration within the muscle, we aim to simplify this regeneration process by assuming that there is one process within the mitochondria regenerating ATP. While there are aerobic, with carbohydrate and fat dependency, and anaerobic processes, we will assume that all of the contractions simulated are within the aerobic regime and ignore the difference with between carbohydrates and fat. Given this restriction, we assume that the contraction intensity is such that the full supply of PCr within the muscle fibre is never fully depleted.

The regeneration of ATP can be described as a system of equations as follows. Let c_{pcr} be the concentration of PCr within the muscle and let a_m be the activation level of mitochondria (unitless, used as a proxy to account for a delay in mitochondrial activation [PTY93]). The system of equations can then be given as follows

$$\dot{c}_{pcr} = -f_{atp}(t) + k_{rec,pcr} a_m (c_{pcr,max} - c_{pcr}) \quad (1)$$

$$\dot{a}_m = k_m (c_{pcr,max} - c_{pcr}) \quad (2)$$

$$c_{pcr}(t = 0) = c_{pcr,max} \quad (3)$$

$$a_m(t = 0) = 0. \quad (4)$$

Here f_{atp} gives the rate of ATP consumed by the muscle contraction. $k_{pcr,rec}$, and k_{delay} are rate constants associated with the recovery of PCr and the delay in mitochondrial activation. **Ryan's comment(s): The delay in mitochondrial activation is important for repeated stimulations, without it there is less of a delay in recovery of pcr.**

While there is physiological intuition behind these parameters, there are not clear rate constants that can be measured and implemented into the equations; thus, these parameters are optimised based on data

at the fibre level from [PTY93]. This is done by minimising the difference between the experimental PCr concentrations and the model predictions

$$\min_{\mathbf{x}=(k_{rec,pcr},k_d)\in\mathcal{S}} \frac{\|c_{pcr,exp}(t_i) - c_{pcr,model}(t_i;\mathbf{x})\|_{L^2}}{\|c_{pcr,exp}(t_i)\|_{L^2}} \quad (5)$$

where we define $\mathcal{S}$ as the bounded set of parameter values

$$\mathcal{S} := \{\mathbf{x} = (x_1, x_2) : x_1, x_2 \in \mathbb{R} \cap [0, 1]\}. \quad (6)$$

The data used to fit the model was taken from Figure 2 [PTY93], which includes data on the time course of the PCr concentrations. Note that the time course of the PCr concentrations were only available at a 4s temporal resolution. The data is shown for a 9s isometric contraction, with a heat rate (given in Table 1 [PTY93]) of $\approx 64 \text{ mJ}(g \text{ of dry wt})^{-1} s^{-1}$. Assuming a ratio of 1 g dry weight per 5 grams of wet weight we have $\dot{Q}_i \approx 12.8 \times 10^{-3} J(g s)^{-1}$ (denoting g as g of wet wt for the remainder of this document). Assuming ATP molecules have $\Delta H_{ATP} = 60 \times 10^3 J \text{ mol}^{-1}$, we can compute the rate of ATP consumption to perform the contraction as

$$f_{ATP}(t) = \frac{\dot{Q}_i}{\Delta H_{ATP}} = 0.213 \times 10^{-6} \frac{\text{mol}}{g s}. \quad (7)$$

The optimal parameter values were determined to be $k_{rec,pcr} = 0.0214$ and $k_d = 0.0367$ (Figure 1).

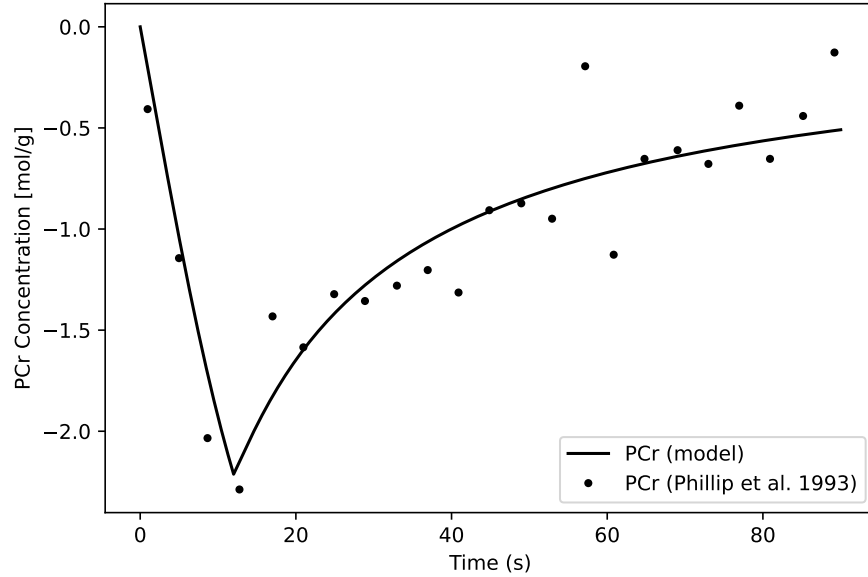


Figure 1: Model fit to experimental data from [PTY93]. Change in phosphate concentrations from resting are shown here with the the first 9s being the contraction with the remaining time being the recovery process.

2.2 Recovery heat

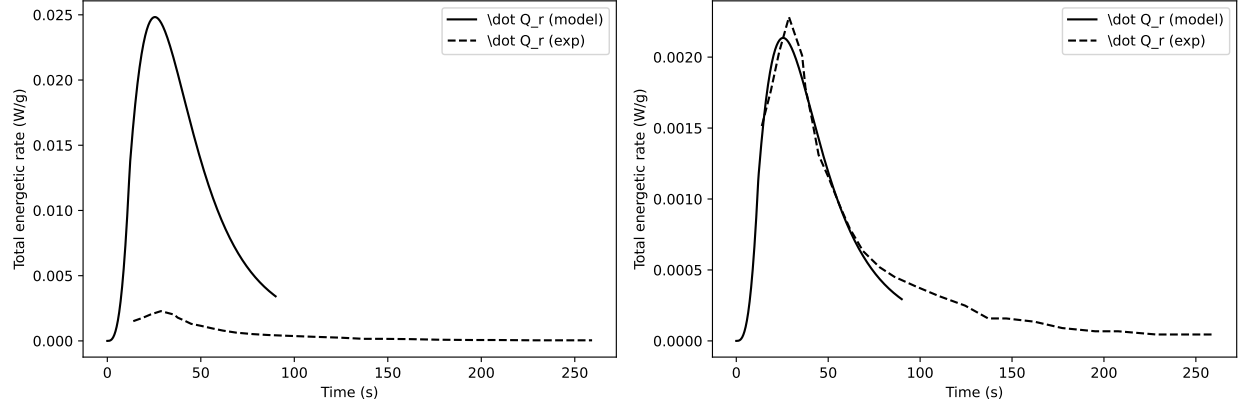
We compute the recovery heat as a constant proportional to the rate of PCr regeneration. This can be obtained using Equation 1, leaving us with the recovery heat rate

$$\dot{Q}_r = r_r k_{rec,pcr} a_m (c_{pcr,max} - c_{pcr}). \quad (8)$$

We can define recovery heat constant r_r (in units of $J(mol)^{-1}$) in two possible ways. One is based on the enthalpy of glucose ($\Delta H_g = -2802 kJ mol^{-1}$) and the corresponding ATP yield of 36 to 40 ATP. Assuming the difference in enthalpy from glucose to ATP is released as heat during the aerobic processes, we have

$$r_r = \Delta H_g - 38 \Delta H_{atp} = 522 \times 10^3 \frac{J}{mol}. \quad (9)$$

This equation results in an over prediction of the energetic rates (**Ryan’s comment(s): Maybe this is an error in my calculation of this constant, since I feel like this should work**). Another method to determine the constant r_r is to fit the energetic rates to the data from [PTY93], as the time course of the heat rate is given post contraction. Doing this results in an $r_r = 44.8 \times 10^3 J mol^{-1}$. The model and experimental energetic rates are shown in Figure 2. Even though the constant is optimised, it is important to note that we see the same time course of recovery heat rate (not optimised). At this point all of the constants are calculated based on experiments at 20°C.



(a) Recovery energetic rates with r_r calculated based on Equation 9. (b) Recovery energetic rates with r_r optimised to the experimental data.

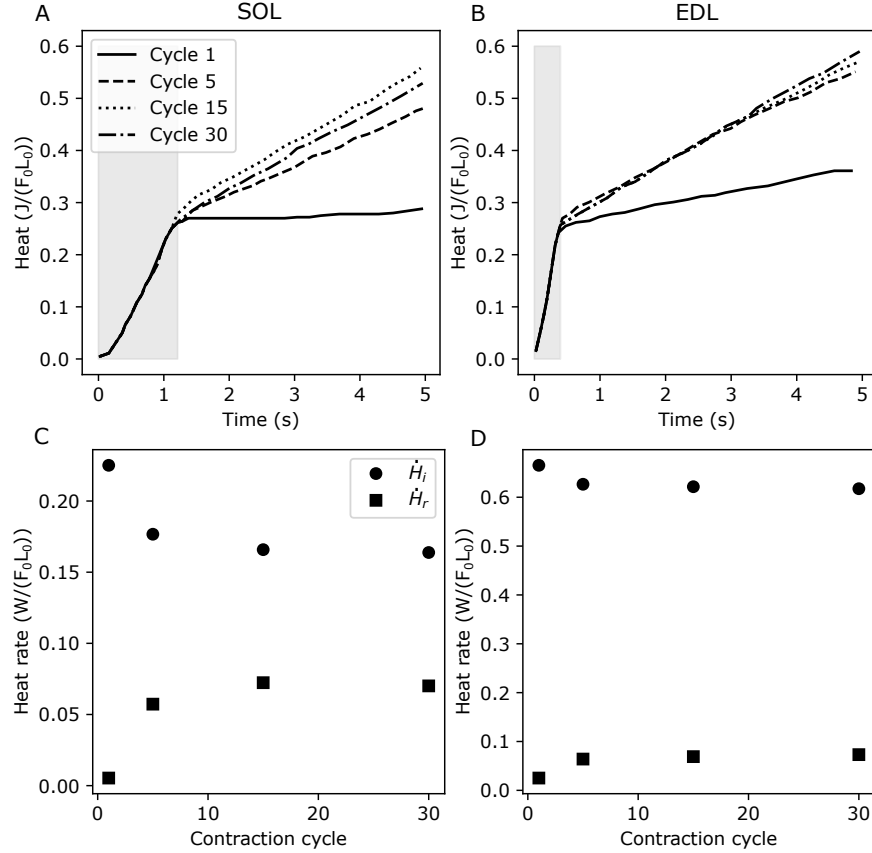
Figure 2: Comparison of predicted recovery energetic rates to experimental values from [PTY93].

3 Numerical experiments

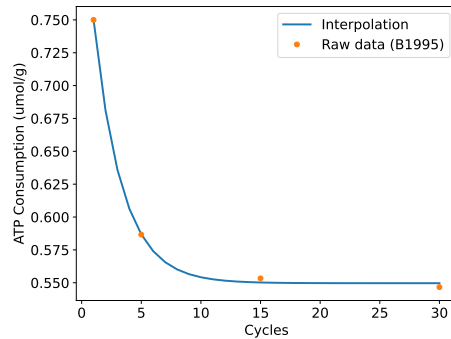
3.1 Comparison to [BAG95]

Simulations were performed to reproduce the experimental setup from [BAG95]. As the data from [PTY93] was taken from rat soleus, we compare to the slow fibre bundles from mouse soleus from [BAG95]. The protocol consisted of isometric contractions with a cycle length of 5 seconds and a duty cycle of 0.16 (fraction of the time active during the cycle), which corresponded to a 0.8s with 4.2s rest. This was repeated for 30

cycles. The experimental heat rates were converted from units of $W (F_0 l_0)^{-1}$ to $W g^{-1}$, then converted to a rate of ATP use in $\mu mol(s g)^{-1}$. The experimental values are shown in **Figure 3** with a fitted exponential function used to extrapolate to the intermediate cycles. The conversion of parameter values is shown in **section 4**. **Ryan's comment(s): I think it makes sense to scale the recovery rate parameter r_r by $F_0 l_0$ instead of by mass? I have not spent the time to figure out how to do this yet. I do not believe all the necessary values were reported in [PTY93].**



(a) Experimental heat and heat rates from [BAG95]. Taken from the JEB Commentary.



(b) Interpolated initial ATP usage.

Figure 3: Experimental heat rates from [BAG95]

The model is able to reproduce some of the time dependent behaviour of the experimentally observed recovery heat rates (**Figure 4**, c.f. **Figure 3**), but does not capture the same ratio of the recovery heat to

initial heat rate. Experimentally, the recovery heat is 43 % of the initial heat rate, while the model only predicted a recovery heat rate of 14% of the initial heat rate (Figure 4, note the units are not the same as in Figure 3). This discrepancy is likely a result of an incorrect parameter value r_r used for the recovery heat rates; however, the model is able to capture the time-dependence of the recovery heat rate with repeated cycles. **Ryan’s comment(s): The r_r parameter is calculated based on rat soleus muscle and is then used to predict heat rates from mouse soleus, could this be an issue? Temperatures were also higher in the [BAG95] at 25°C compared to 20°C in the [PTY93].**

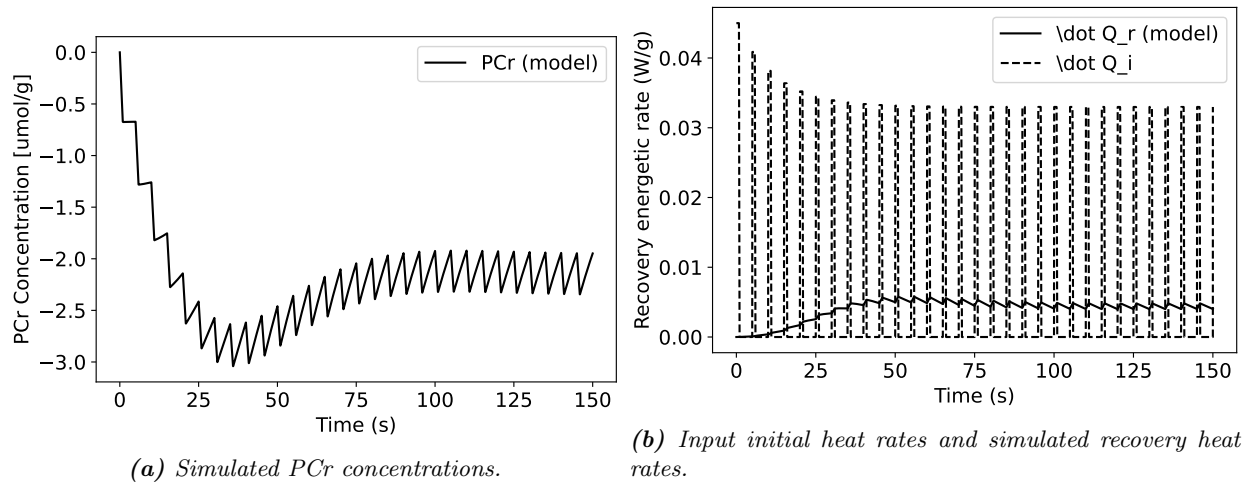


Figure 4: Reproducing results from [BAG95]. Simulations include repeated contractions with a 5 second cycle with a contraction of 0.8s. The initial heat rate from the contraction was variable (decreased with time), which is captured in the initial heat rates given as input to the model.

References

- [PTY93] S. K. Phillips, M. Takei, and K. Yamada. “The Time Course of Phosphate Metabolites and Intracellular pH Using ^{31}P NMR Compared to Recovery Heat in Rat Soleus Muscle.” In: *The Journal of Physiology* 460.1 (Jan. 1993), pp. 693–704. ISSN: 14697793. DOI: [10.1113/JPHYSIOL.1993.SP019494](https://doi.org/10.1113/JPHYSIOL.1993.SP019494). (Visited on 11/14/2024).
- [BAG95] C. J. Barclay, P. D. Arnold, and C. L. Gibbs. “Fatigue and Heat Production in Repeated Contractions of Mouse Skeletal Muscle.” In: *The Journal of Physiology* 488.3 (Nov. 1995), pp. 741–752. ISSN: 1469-7793. DOI: [10.1113/JPHYSIOL.1995.SP021005](https://doi.org/10.1113/JPHYSIOL.1995.SP021005). (Visited on 05/03/2024).

4 Appendix

4.1 Code to scale parameters

```
# Constants from Barclay 1995
sigma_0 = 2e5          # Pa
mass = 0.51e-3         # g
rho = 1e6              # g/m^3
l_0 = 9.7e-3          # m

G_atp = 60e3           # J/mol Free energy of ATP

dHidt_f0l0 = 0.225     # W/F_0/l_0, Initial heat rate

####
# Convert the heat rates to W/g and get ATP consumption rate

# Compute CSA
A_csa = mass / rho / l_0 # m^2
print(f'A_csa = {A_csa} m^2')

# Compute maximum isometric force
F_0 = sigma_0 * A_csa # N
print(f'F_0 = {F_0} N')

# Compute the heat rate in W/g
dHidt = dHidt_f0l0 * F_0 * l_0 / mass # W/g
print(f'dHidt = {dHidt} W/g')

# Compute the rate of atp consumption
dATPdt = dHidt / G_atp * 10**6 # umol/g
print(f'dATPdt = {dATPdt} umol/g')
```

4.2 Model implementation (for the [BAG95] simulations)

```
'''
This is a code to simulate bioenergetics of muscle contraction
Within this code the modelling is limited to the amount of PCr within the muscle,
based on a prescribed use of ATP.
The full contraction is not simulated within this code

This version is used to compare model predictions to those from Barclay 1995.

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'''

# Import
import numpy as np
import matplotlib.pyplot as plt
plt.rcParams['font.size'] = 14
```

```

133 import matplotlib.cm as cmap
134 import sys
135 sys.path.append('./')
136
137 # Parameters
138 params = {
139     # Time parameters
140     't_start': 0, # s
141     't_end': 30 * 5, # s, 30 cycles at 5 seconds each
142
143     # Energetics parameters
144     'Energetics_model':{
145         # Define parameters for the ODE
146         # .... In this code they are used as initial values for the optimization
147         'k_recpcr': 0.0214, # 1/s
148         'k_m_1': 0.0367, # 1/s
149
150         # Energetic constant to predict energetic rates
151         # 'r_r': 522e-3, # J umol-1, Calculated based on the glycogen and atp enthalpys
152         'r_r': 0.04489659, # J umol-1, Optimized to experimental data (Phillips et al. 1993
153         )
154     }
155 }
156
157 ###
158 # Create a class for the ode solver
159 class Bioenergetics():
160     def __init__(self, k_recpcr, k_m):
161         # Initialize the main parameters
162         self.k_recpcr = k_recpcr
163         self.k_m = k_m
164
165         # Counter for the cycles
166         self.cycle_count = 1 # NOTE: starts at 1
167
168     # Compute the recovery of ATP
169     def pcrRecoveryRate(self, t, y):
170         '''
171         Function of time and current pcr concentration
172         '''
173         pcr, Ma = y
174
175         # New model 2a
176         dpcrdt = - self.ATP(t) + self.k_recpcr * Ma * (pcr)
177         dMadt = self.k_m * (pcr - Ma)
178
179         # Return rate of recovery
180         return (dpcrdt, dMadt)
181
182     def ATP(self, t):
183         '''
184         Function defining ATP use during the contraction
185
186         Modified to simulate the contractions from Barclay et al. 1995

```

```

187     '''
188     # Constant atp_peak [we use variable atp usage now... see below]
189     # atp_peak = 0.75 # umol/s/(g wet wt) [computed using computeParametersBarclay1995.
190                                     py]
191
192     tstimend = 0.8 # s, Length of stimulation (B1995)
193     trampend = 1
194
195     # Normalize time
196     t_cycle_length = 5 # s, Length of the cycle
197     t_cycle = t%t_cycle_length
198
199     # Use variable ATP usage
200     def f(x, a, b, c, d):
201         '''
202         Function as defined in computeParametersBarclay1995.py
203         *** Ensure its the same if any adjustments are made (e.g. fibre-type) ***
204         '''
205         return a * np.exp(b * x - c) + d
206     popt = np.array((26.31884038, -0.42107639, 4.45702316, 0.54965355))
207     cycle_count = np.floor(t/t_cycle_length) + 1
208     # Compute the atp usage based on the cycle number
209     atp_peak = f(cycle_count, *popt) # umol/s/(g wet wt) [computed using
210                                     computeParametersBarclay1995.py]
211
212     return atp_peak * (t_cycle < tstimend) + atp_peak * 0.5 * (np.sin((np.pi*(t_cycle-
213                                     tstimend) / (trampend-tstimend) + np.
214                                     pi/2)) + 1) * (t_cycle > tstimend) * (
215                                     t_cycle < trampend)
216
217     def solveODE(self, t_start, t_end):
218         '''
219         Function to solve the ode
220         '''
221         # Initial conditions for ode
222         PCr_initial = 0.0 # Initial d[PCr] in mM
223         activation_initial = 0.0 # Initial mitochondrial activation
224         y0 = [PCr_initial, activation_initial]
225
226         # Solve the ode
227         from scipy.integrate import odeint, solve_ivp
228         tspan = (t_start, t_end)
229
230         # Solve the IVP
231         sol = solve_ivp(self.pcrRecoveryRate, tspan, y0, max_step = 0.01)
232
233         return sol
234
235     def computeEnergetics(self, pcr, Ma, r_r):
236         '''
237         Function to compute the energetic rate of the recovery process
238         '''
239         return r_r * self.k_recpcr * Ma * pcr
240

```



```

241 # Run Bioenergetics model
242 model = Bioenergetics(params['Energetics_model']['k_recpcr'], params['Energetics_model']['
243                               k_m_1'])
244 sol = model.solveODE(params['t_start'], params['t_end']) # Solve the IVP
245 PCr, activation = sol.y
246 Q_r = model.computeEnergetics(PCr, activation, params['Energetics_model']['r_r']) # Compute
247                               the recovery energetic rate
248
249 # Plot the comparison between PCr experimental and modelled
250 fig, ax = plt.subplots(figsize = (6,4), layout = 'constrained')
251 ax.plot(sol.t, PCr, label = 'PCr (model)', color = 'k')
252 # ax.plot(t_exp, pcr_exp, label = 'PCr (Phillip et al. 1993)', ls = 'None', marker = '.',
253                               color = 'k')
254 ax.set_xlabel('Time (s)')
255 ax.set_ylabel('PCr Concentration [umol/g]')
256 plt.legend()
257 plt.savefig('./Figures/B1995PCr.pdf')
258
259 # Plot the energetic rates (experimental and predicted rates)
260 fig, ax = plt.subplots(figsize = (6,4), layout = 'constrained')
261 ax.plot(sol.t, Q_r, label = '\dot Q_r (model)', color = 'k')
262 ax.plot(sol.t, model.ATP(sol.t) * 60e3 / 10**6, label = '\dot Q_i', color = 'k', ls = '
263                               dashed')
264 ax.set_xlabel('Time (s)')
265 ax.set_ylabel('Recovery energetic rate (W/g)')
266 plt.legend()
267 plt.savefig('./Figures/B1995Energetics.pdf')
268
269
270 plt.show()
271

```